Coping with NP-completeness: Introduction

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Advanced Algorithms and Complexity
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- Your boss asked you to implement a program that solves efficiently a certain search problem

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- If you are lucky enough, the problem can be solved by some known technique like dynamic programming, linear programming, flows (though it is usually still not immediate to notice this)
- Alas, this happens rarely

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Michael R. Garey and David S. Johnson.
Computers and Intractability: A Guide to the Theory of NP-Completeness. 1979.

Perhaps there is just no efficient algorithm for your search problem.

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- Instead of showing that there is no efficient algorithm for your program, you show that it is one of the hardest search problems
- That is, you show that your problem is **NP**-complete

“I can’t find an efficient algorithm, but neither can all these famous people!”

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Keep your head up!

It is just the beginning of a fascinating adventure!

Next Parts

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	poly time	optimal solution	all cases
special cases	✓	✓	✗
approximation algorithms	✓	✗	✓
exact algorithms	✗	✓	✓